

STUDENT RECORD MANAGEMENT SYSTEM

One-Page Project Summary | CSE Micro Project | Session 2026–2027

Project Overview

A simple console-based application developed in C to digitally store, retrieve, search, and delete student records. It replaces paper-based record keeping with computer memory and permanent file storage, making basic record operations faster and easier to manage.

Key Features

Add Student	Enter and save roll number, name, branch, and CGPA.
View Records	Display all stored student records in a formatted table.
Search Student	Find a student using the roll number.
Delete Student	Remove a record using a temporary-file based deletion method.
File Storage	Store records permanently in the binary file students.dat.
Menu-Driven Interface	Simple switch-case menu for easy console operation.
CRUD Operations	Supports Create, Read, and Delete; the report identifies the overall system as CRUD.

Technologies & Concepts Used

Programming Language	C
Data Structure	struct Student (roll number, name, branch, CGPA)
File Handling	fopen(), fwrite(), fread(), fclose(), remove(), rename()
String Handling	<string.h>, including strcmp()
Control Logic	switch-case and while loops
Compiler	GCC / MinGW
Development Environment	Dev C++, Code::Blocks, or Visual Studio Code
Data File	students.dat (binary file)

Objectives & Future Scope

The project aims to build a fast, reliable, easy-to-use digital record system and provide practical experience with C structures and file handling. Future improvements mentioned in the report include an admin login/password, CSV export, and automatic sorting by roll number or CGPA.

Team

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